Kernix 0.4.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

13 workloads

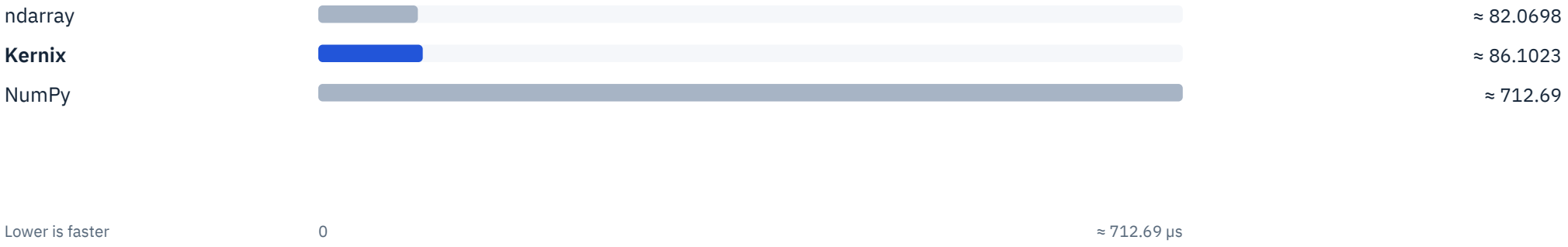
38 observations

Source precision retained

Matrix multiplication

Batch-mean median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Historical medians of 15 batch means. Recorded p95 is the batch maximum.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.4.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Matrix multiplication

Batch-mean median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product

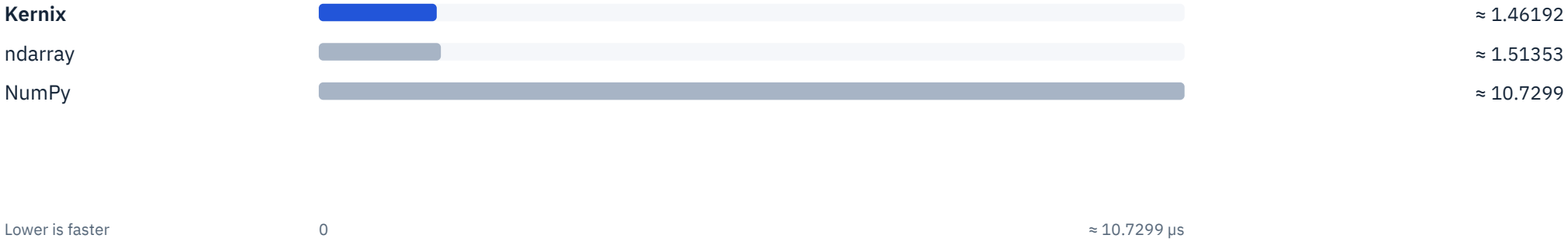
Kernix 0.4.0 / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Matrix multiplication · reused output

Batch-mean median

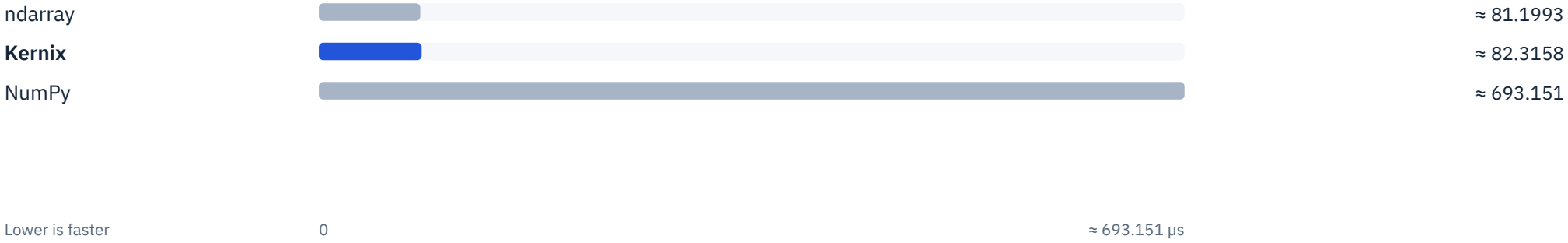
32 × 32 · 32 × 32 · f64 · 1 worker · reused output



Matrix multiplication · reused output

Batch-mean median

128 × 128 · 128 × 128 · f64 · 1 worker · reused output



Kernix 0.4.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Cholesky factorization

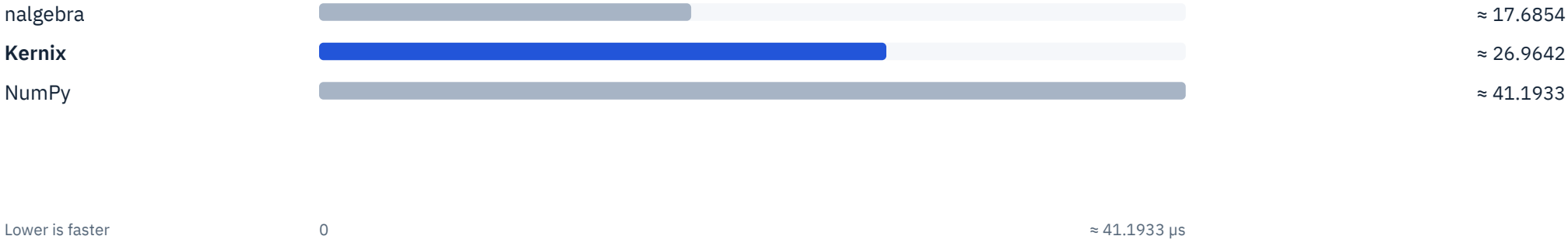
Batch-mean median

64 × 64 · f64 · 1 worker · allocated output

Linear system solve

Batch-mean median

64 × 64 · f64 · 1 worker · allocated output



Kernix 0.4.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Thin QR

Batch-mean median

96 × 32 · f64 · 1 worker · allocated output

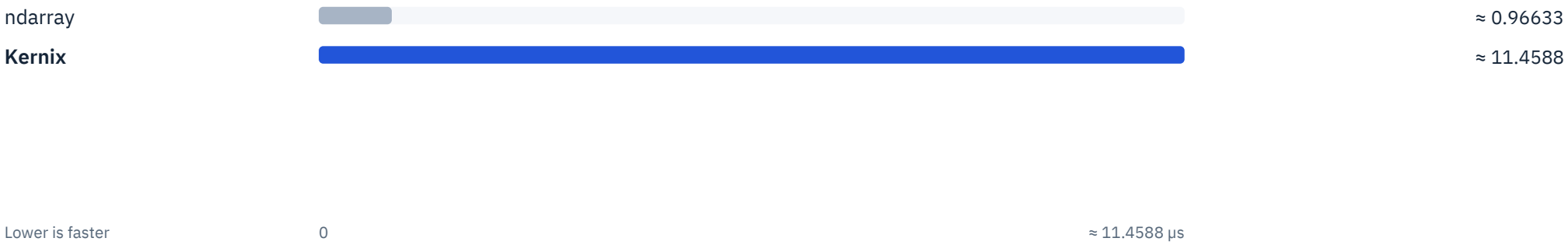
Kernix 0.4.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Dense training step

Batch-mean median

32 × 16 × 8 · f64 · Graph-based Kernix vs hand-derived comparator updates; API costs differ



Kernix 0.4.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Elementwise addition

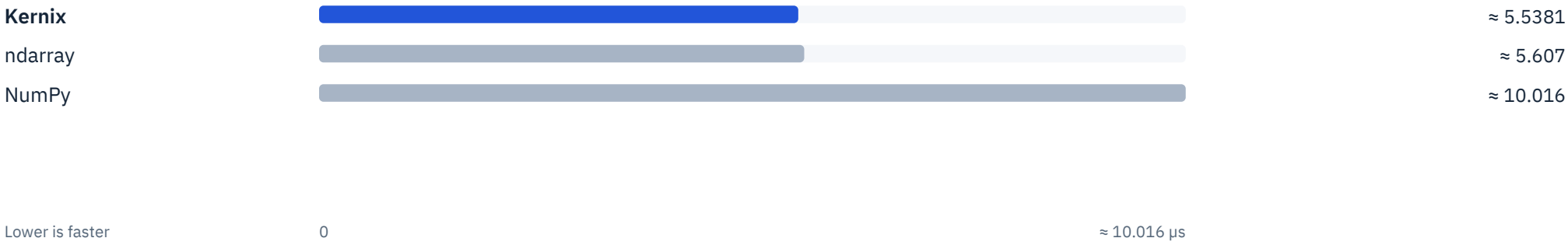
Batch-mean median

65,536 elements · f64 · 1 worker · allocated output

Sum reduction

Batch-mean median

65,536 elements · f64 · 1 worker · scalar output



Kernix 0.4.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Batch-mean median

Kernix 0.4.0 / Vision workloads

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: cblas

RGB resize & normalization

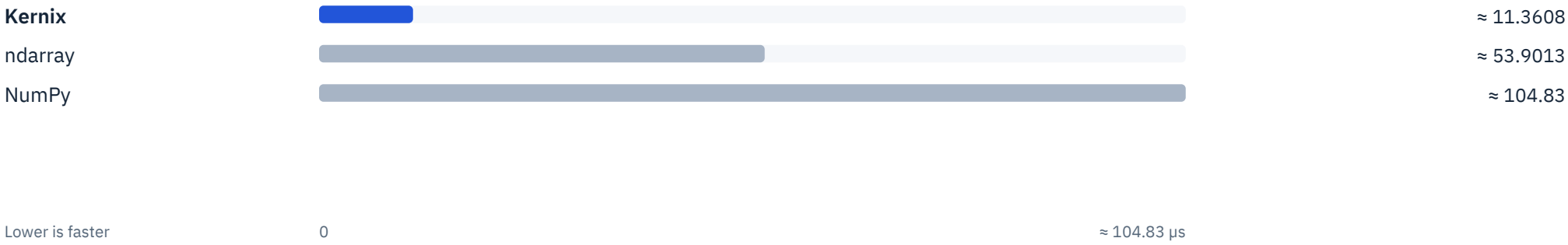
Batch-mean median

1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32 · 1 worker · allocated output · comparator implementations differ

2D convolution

Batch-mean median

1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32 · 1 worker · allocated output · comparator implementations differ



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1600.2428470844268	1.6002428470844268
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1416.7458491528305	1.4167458491528305
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	10639.569787234042	10.639569787234042
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	86102.32989690722	86.10232989690722
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	82069.83934426229	82.06983934426229
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	712689.7777777778	712.6897777777778
Matrix multiplication · reused output / Kernix 32 × 32 · 32 × 32 · f64	1461.9173147769136	1.4619173147769136
Matrix multiplication · reused output / ndarray 32 × 32 · 32 × 32 · f64	1513.5317229688824	1.5135317229688824
Matrix multiplication · reused output / NumPy 32 × 32 · 32 × 32 · f64	10729.9141999142	10.7299141999142
Matrix multiplication · reused output / Kernix 128 × 128 · 128 × 128 · f64	82315.77302631579	82.31577302631579

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · reused output / ndarray 128 × 128 · 128 × 128 · f64	81199.30844155845	81.19930844155845
Matrix multiplication · reused output / NumPy 128 × 128 · 128 × 128 · f64	693150.5675675676	693.1505675675676
Cholesky factorization / Kernix 64 × 64 · f64	23374.6476635514	23.3746476635514
Cholesky factorization / nalgebra 64 × 64 · f64	10564.963244613435	10.564963244613435
Cholesky factorization / NumPy 64 × 64 · f64	26421.718057022175	26.421718057022175
Linear system solve / Kernix 64 × 64 · f64	26964.24676724138	26.96424676724138
Linear system solve / nalgebra 64 × 64 · f64	17685.366336633662	17.685366336633662
Linear system solve / NumPy 64 × 64 · f64	41193.26194398682	41.19326194398682
Thin QR / Kernix 96 × 32 · f64	88613.39929328622	88.61339929328622
Thin QR / nalgebra 96 × 32 · f64	32263.832258064514	32.263832258064514

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 96 × 32 · f64	79375.3746031746	79.3753746031746
Dense training step / Kernix 32 × 16 × 8 · f64	11458.840971585701	11.458840971585701
Dense training step / ndarray 32 × 16 × 8 · f64	966.3295068027211	0.9663295068027211
Elementwise addition / Kernix 65,536 elements · f64	21233.55432937182	21.23355432937182
Elementwise addition / ndarray 65,536 elements · f64	22836.65296803653	22.83665296803653
Elementwise addition / NumPy 65,536 elements · f64	22884.931381518756	22.884931381518756
Sum reduction / Kernix 65,536 elements · f64	5538.095459579181	5.538095459579181
Sum reduction / ndarray 65,536 elements · f64	5607.004709576138	5.607004709576138
Sum reduction / NumPy 65,536 elements · f64	10016.018822587104	10.016018822587104
Row softmax / Kernix 256 × 64 · f64	77015.27076923077	77.01527076923077

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / ndarray 256 × 64 · f64	69555.83888888889	69.55583888888889
Row softmax / NumPy 256 × 64 · f64	94432.09056603773	94.43209056603773
RGB resize & normalization / Kernix 1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32	438953.1379310345	438.9531379310345
RGB resize & normalization / ndarray 1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32	685549.3513513514	685.5493513513514
RGB resize & normalization / NumPy 1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32	3994365.4285714286	3994.3654285714286
2D convolution / Kernix 1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32	11360.770104497955	11.360770104497955
2D convolution / ndarray 1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32	53901.27370689655	53.90127370689655
2D convolution / NumPy 1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32	104830.2510460251	104.8302510460251